

Worksheet 4

Chapter 4.1 The basics of probability

Solve the following questions:

4.3 If two customers are asked to list their choice of ice cream flavor from among vanilla, chocolate, and strawberry, list the sample space showing the possible outcomes.

4.4. Use a tree diagram to list the sample space for the number of movies rented by three customers at a video store where customers are allowed to rent one, two, or three movies (assuming that each customer rents at least one movie).

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4.8 The results of a census of **2,500** employees of a midsized company with 401 (k) retirement accounts are as follows:

Account Balance (to nearest \$)	Male	Female
Less than < \$ 25,000	635	495
\$ 25,000 to \$ 49,000	185	210
\$ 50,000 to \$ 99,999	515	260
More than >\$ 100,000	155	45

Suppose researchers are going **to sample employees** from the company for further study.

- What is the probability that a randomly selected employee will be a female?
- What is the probability that a randomly selected employee will have a 401(k) account balance, of between \$25000 and \$49999?
- Compute the probability that a randomly selected employee will be a female with an account balance between \$50,000 and \$99,999.

4.9 Cross County Bicycles makes two **mountain bike models**, the XB-50 and the YZ -99, in three distinct colors. The following table shows the production volumes for 1st week:

Model	Color		
	Blue	Brown	White
XB-50	302	105	200
YZ-99	40	205	130

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- a. What is the probability that a mountain bike is brown?
- b. What is the probability that the mountain bike is a YZ-99?
- c. What is the probability that a randomly selected mountain bike is a YZ-99 and brown?
- d. Suppose a mountain bike is chosen at random. Consider the following two events:
the **event that model YZ-99 is chosen** and the **event that a white product is chosen**.
Are these two events mutually exclusive? Explain.

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Chapter 4.2: The Rules of Probability

4-26 A study was conducted in a school on how students travel to school. Following are the data collected for three methods students travel to school:

Method	Number of Students
Carpool	35
Drive	14
Public Transport	47

- a) Construct a relative frequency table for the data
- b) What is the probability the students is not driving to school?
- c) What is the probability that student use either carpool or drives to school?

- 4-27.** The Jack In The Box franchise in Bangor, Maine, has determined that the chance a customer will order a soft drink is 0.90. The probability that a customer will order a hamburger is 0.60. The probability that a customer will order french fries is 0.50.
- a. If a customer places an order, what is the probability that the order will include a soft drink and no fries if these two events are independent?
 - b. The restaurant has also determined that if a customer orders a hamburger, the probability the customer will also order fries is 0.80. Determine the probability that the order will include a hamburger and fries.

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4.31: Three events occur with probabilities:

$P(E1) = 0.35$, $P(E2) = 0.15$, $P(E3) = 0.40$. If the event B occurs, the probability becomes $P(E1|B) = 0.25$, $P(B) = 0.30$

a) Compute $P(E1 \text{ and } B)$

b) Compute $P(E1 \text{ or } B)$

c) Assume $E1$, $E2$, $E3$ are independent event. Compute $P(E1 \text{ and } E2 \text{ and } E3)$

4.32. Hubble Construction Company has submitted a bid on a state government project that is to be funded by the federal government's stimulus money in Arizona. The price of the bid was predetermined in the bid specifications. The contract is to be awarded on the basis of a blind drawing from those who have bid. Five other companies have also submitted bids. What is the probability of the Hubble Construction Company winning the bid?

Suppose a quality manager for Dell Computers has collected the following data on the quality status of disk drives by supplier. She inspected a total of 700 disk drives.

Supplier	Drive Status	
	Working	Defective
Company A	120	10
Company B	180	15
Company C	50	5
Company D	300	20

What is the probability of a defective disk drive being received by the computer company?

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2. A tyre manufacturing company kept a record of the distance covered before a tyre needed to be replaced. The table shows the results of 1000 cases.

Distance(in km)	Less than 4000	4000 to 9000	9001 to 14000	More than 14000
Frequency	20	210	325	445

If a tyre is bought from this company, what is the probability that :

(i) it has to be substituted before 4000 km is covered?

(ii) it will last more than 9000 km?

(iii) it has to be replaced after 4000 km and 14000 km is covered by it?

3. The percentage of marks obtained by a student in the monthly tests are given below:

Test	1	2	3	4	5
Percentage of marks obtained	69	71	73	68	74

Based on the above table, find the probability of students getting more than 70% marks in a test.

9. A coin is tossed three times, consider the following events.

P: 'No head appears',

Q: 'Exactly one head appears' and

R: 'At Least two heads appear'.

Example 3: There are 5 green 7 red balls. Two balls are selected one by one without replacement. Find the probability that first is green and second is red.

Example 4: What is the probability of getting a sum of 7 when two dice are thrown?

Example 1: A coin is thrown 3 times .what is the probability that atleast one head is obtained?